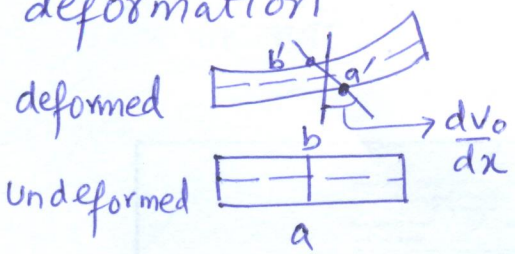
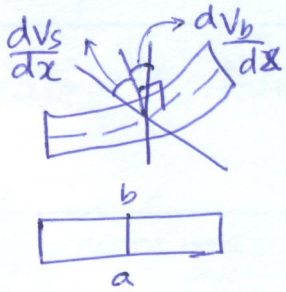
Difference between Euler-Bernoulli beam Theory and Timoshinko beam theory :-

In Euler-Bernoulli beam Theory, shear deformations are neglected and plane sections remain plane and Normal to the longitudinal axis.

In Timoshinko beam theory, plane sections still remain plane but are no longer normal to the longitudinal axis. The difference between the normal to the longitudinal axis and the plane section rotation is the shear deformation

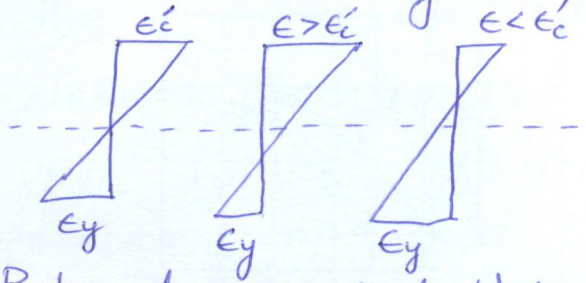




→ Conventional Assumptions

- Perfect bond
- Stress Resultants and external actions are in equilibrium
- $\sigma\epsilon$ relation of Material is known
- Plane sections remain plane
- The concrete limiting strain is defined
- Steel perfectly Elasto plastic. $\epsilon_c = 0.003$

→ Concrete crushing and yielding of reinforcing bars take place simultaneously.



Balanced over-Reinforced Under-Reinforced

→ $M_{rc}\ (f_c', f_y, b, d)$

→ M_{cr} = Measure of the capacity of concrete in compression to resist moment.
It also ensures some ductility by forcing failure in tension.

→ $M_u = \phi M_n$; $M_n = A_s f_y \left(d - \frac{a}{2}\right)$ for singly Rectangular

→ For flexure: $A_{s,min} = \frac{3\sqrt{f_c'}}{f_y} b_w d \geq \frac{200 b_w d}{f_y}$ [for Rectangular + T with flange in Comp]

$A_{s,min} = \frac{6\sqrt{f_c'}}{f_y} bwd$ [for T with flange in Tension]

$A_{s_{max}}$:- $\rho_{max} = \rho' + 0.75\rho_b$

$\rho' = \frac{A_{sc}}{bd}$

→ Cracking depends on distribution of rebars in the tension zone and on steel stress.

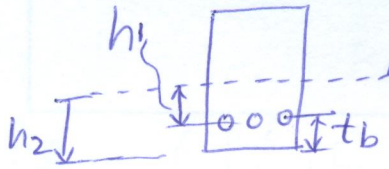
$$w = 0.076 \beta f_s \sqrt[3]{d_c A}$$

At extreme fiber :-

$$w_{max} = \left[0.076 \sqrt[3]{t_b A} \frac{h_2}{h_1} f_s \right] \times 10^{-6} \text{ in}$$

t_b = distance from extreme tension fiber to the center of the adjacent bar (in)

A = average effective area of concrete around each bar (Ae/n)



$f_s = 0.6 f_y$

→ $M_u, f_c', f_y \longrightarrow M_{rc}$

$M_{rc} > M_u$ ← NO — Doubly A_{st}, A_{sc} → Check $A_{st,max}$ → Use min ← NO — Check $A_{st,min}$

$M_{rc} > M_u$ → yes → Singly → Check $A_{st,min}$ → yes, OK → Determine Layout → Complete

Unified Theory for Concrete Design

→ unifying beams + cols
RC + Prestressed
WSD + USD
Cross-section types
Stress - Strain models

"Partially prestressed Composite Section Subjected to arbitrary bending"

→ If we have general $\sigma\epsilon$ curve
- → all materials are same
- → Concept of material doesn't exist now.

Assume Strain Profile → angle of NA, Depth of NA, max strain
↓
determine stress at each level
↓
determine stress in each Component from corresponding $\sigma\epsilon$ curve
↓
Calculate Resultant for each Component
↓
Calculate Total stress Resultant by summing components.

Governing Equations:

$$N_z = \phi_1\left[\frac{1}{\gamma_1}\int_x\int_y \sigma(x,y)\,dx\,dy \text{ -------} + \frac{1}{\gamma_2}\sum_{i=1}^{n} A_i\sigma_i(x,y) \text{---}\right]$$

$$M_x = \phi_2\left[\frac{1}{\gamma_1}\int_x\int_y \sigma(x,y)\,dx.dy.y \text{ ---} + \frac{1}{\gamma_2}\sum_{i=1}^{n} A_i\sigma_i(x,y)y_i \text{---}\right]$$

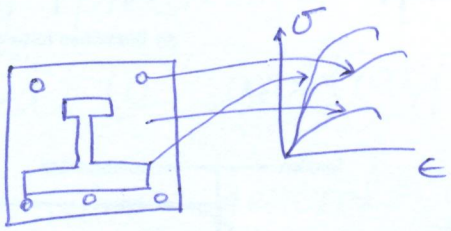
$$M_y = \phi_3\left[\frac{1}{\gamma_1}\int_x\int_y \sigma(x,y)\,dx.dy.x \text{ ----} + \frac{1}{\gamma_2}\sum_{i=1}^{n} A_i\sigma_i(x,y)\cdot x_i\right]$$

These equations cannot be used without specifying reinforcements.

so these are analysis. but can be used for design.

"Understanding Cross Sections"
by
Dr. Naveed Anwar

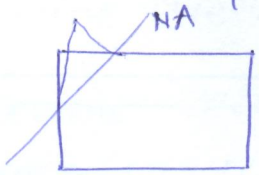
→ Failure Capacity Surface → any combination of N, M_x and M_y if & inside → safe.

→ 

→ Moment shift the load from center.

→ Moment means equivalent eccentric load.

→ Intersection b/w deformed and undeformed Section is neutral plane.



Sudden stress drop to zero if we follow ACI.

→ Stresses at any load level can be calculated using previous page equations and iteration between strain values and neutral axis position.

→ Perfect bond assumption → not required in this because for "any" strain → we have stress.

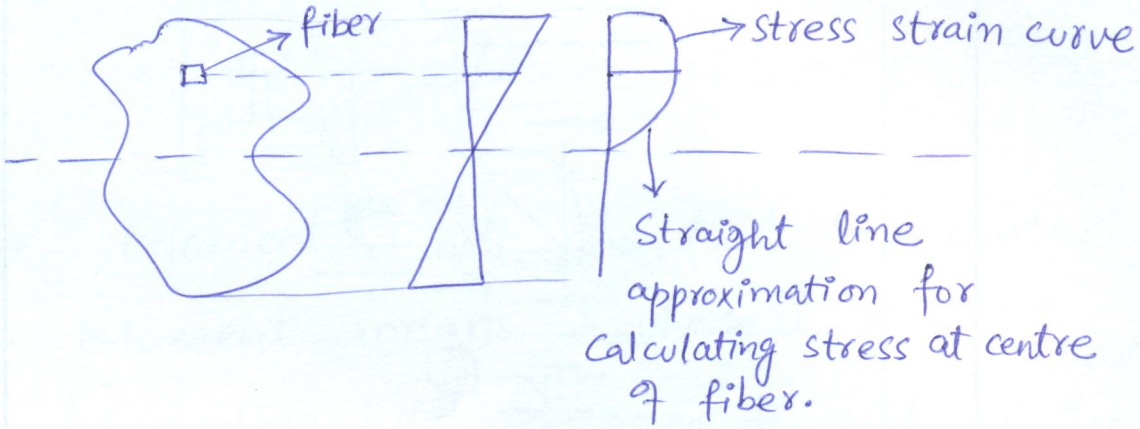
→ Plane sections remain plane → also not required.

Fiber Model

→ We divide our cross-section in to small areas → fibers

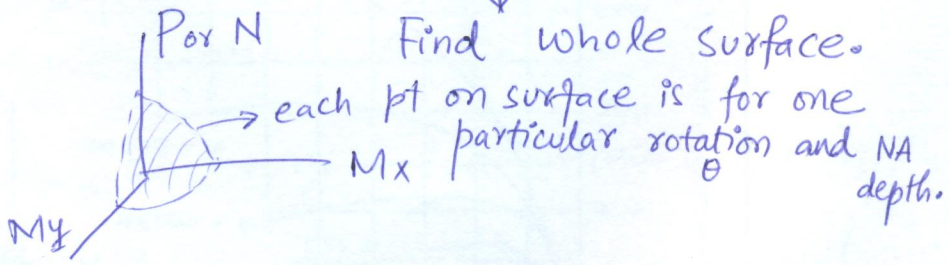
→ Change $\int$ in equations in to Σ → do in excel.

→ Descritized curves in local axis

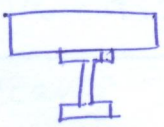


The final stress resultant will be for one particular NA depth.

Chang NA → Find stress Resultants → Generate curves



→ Fiber model → nonlinear by definition because it uses 

→ It can be used to study slip b/w slab and beam. 

→ For steel → problem of local buckling → material cannot reach yield → Modify $\sigma\epsilon$ curve including buckling.

→ Any problem → local buckling, bond slippage → → modify σε curve

either modify strain profile or σε curve.

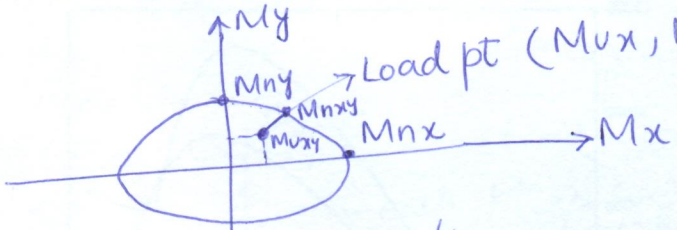
Shape of x-section determines the shape of interaction surface.

→ Capacity surface doesn't give you the precise capacity b/c answer is not unique.

→ Moment sense can be defined by corresponding eccentricity.

→ The axial-flexural capacity of the Cross-section is represented by three stress resultants.

→

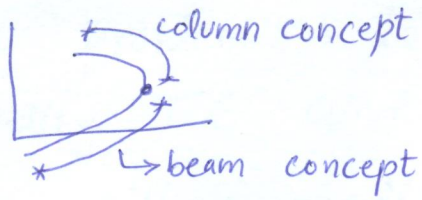


base on Sum of at Ms → PU → $C_r = \left(\frac{M_{ux}}{M_{nx}} + \frac{M_{uy}}{M_{ny}}\right) \leq 1$ @ point

Moment Vector at PU → $C_r = \left(\frac{M_{uxy}}{M_{nxy}}\right) \leq 1$ @ pt

True Capacity Vector in 3D → $C_r = \left|\frac{\overrightarrow{P_u, M_{uxy}}}{\overrightarrow{P_n, M_{nxy}}}\right| \leq 1$

→ Each load comb. gives different FOS So we dont know what is reserve strength.



→ Box section is most effective in torsion.

Best resistance against

→ Torsion ⟶ farthest from centre so → maximum stess lever arm so it concentrates on corners.

→ In Shear → in fact failure is in tension

$v_c = 2\sqrt{f_c'}$ (empirical)

→ In flexure we say concrete take no tension but in shear we say it can take → V_c

→ In BS → no V_c considered so design reinforcement directly.

→ In EQ design → no V_c considered.

→ In the absence of direct compressive stress → one of the principal stress will be in tension so diagonal tension.

→ for shear → only that portion of shear x-section which take shear.

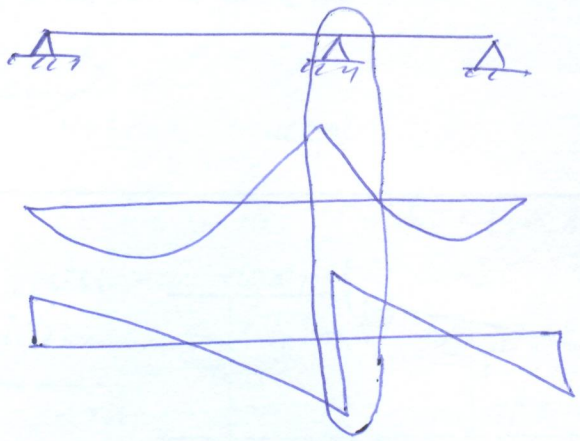
→ Whatever you do in design will be for one point / location.

→ Portion of beam which is in tension has less shear capacity.

→ Torsion + shear → additive in some locations, subtracting on some locations

→ Shear → compounded with axial, M, T.

→ effect of M on shear capacity???



High V, High M

b_w = only web, ignore flanges.

→ Interaction between shear and Torsion
  ↳ Truss mechanism.

→ In Torsion → we need 4 corner bars for Truss to work.

→  Limit "a" to certain value

If c → too much large → brittle failure.

a_{max} → To ensure that steel yields.

If $a > a_{max}$ → Put compression steel.

→ Min reinforcement → To avoid sudden brittle Tension failure.

Max reinforcement → Controlled for compression brittle failure.

→ In design of doubly r/f beam,

$f_{sev} = \frac{2}{3} f_y$

$S_{1,max}$ = smaller of a) $15 \times \frac{40000}{f_s} - 2.5 C_c$

b) $12 \times \frac{40000}{f_s}$

→ For Crackwidth.

→ In beam if M is negative → bottom b
" " +ive → Top b.

→ Three ranges of Shear

$$V_u < V_c$$

$$V_c \leq V_u \leq 0.5V_c$$

$$V_u > V_c$$

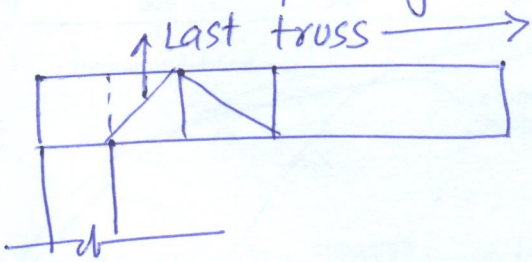
$S \leq \frac{d}{2}$
$S \leq \frac{d}{4}$
→ sudden change in limits
Why not continuous formula.

$\frac{A_v}{S}$ → reinf density

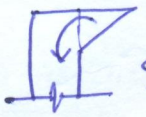
A_v = Total area of all legs.

Increase or decrease 3 things

- Dia
- NO of bars
- Spacing.



→ shear beyond that will be cattered by bracket effect



Shear goes to column.

→ V_s is limited by $8\sqrt{f'_c} b_w d$